A 0.45 Berry–Esseen Bound

Regularized Prawitz Estimates and Finite Certificate Verification

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Abstract

Let $X_1, X_2, \dots$ be independent and identically distributed random variables with mean zero, variance $\sigma^2 > 0$, and finite third absolute moment. We develop a complete analytic and computational route to

$$\sup_{x\in\mathbb{R}} \left| \mathbb{P}\left\{ \frac{X_1 + \cdots + X_n}{\sigma\sqrt{n}} \le x \right\} - \Phi(x) \right| \le 0.45\, \frac{\mathbb{E}|X_1|^3}{\sigma^3\sqrt{n}}.$$

Known inequalities reduce the problem to $2 \le n \le 8123$ and $1 \le \rho \le 712569/589000$, where $\rho = \mathbb{E}|X_1|^3/\sigma^3$. On this rectangle we combine Prawitz's smoothing inequality with a Rademacher comparison and two complementary characteristic-function majorants. Three analytic devices make rigorous interval evaluation stable: a positive partial-fraction expansion for a regularized cotangent, a nonnegative factorization of the Gaussian comparison defect, and an exact endpoint identity for the Prawitz kernel that gives a quadratic tail. The resulting criterion is formulated for cell-by-cell verification with outward-rounded Arb arithmetic. We specify the certificate language, exact partition reconstruction, trust boundary, fail-closed checker, and reproducible execution protocol needed to turn the finite computation into a mathematical proof. The archival run reported in the final section completed this verification.

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1 Introduction

For a centered random variable Y with $\mathbb{E}Y^2 = 1$ and $\rho = \mathbb{E}|Y|^3 < \infty$, write

$$S_n = \frac{Y_1 + \cdots + Y_n}{\sqrt{n}}, \qquad \Delta_n = \sup_{x\in\mathbb{R}} |\mathbb{P}\{S_n \le x\} - \Phi(x)|.$$

The classical Berry–Esseen problem asks for the smallest universal constant C^* such that

$$\Delta_n \le C^* \frac{\rho}{\sqrt{n}}$$

for every n and every admissible distribution.

The best previously published general upper bound is $C^* \le 0.4690$, while Esseen's asymptotic two-point example gives

$$C^* \ge C_E = \frac{\sqrt{10}+3}{6\sqrt{2\pi}} = 0.409732\ldots.$$

Thus the value of C^* remains open; see Shevtsova [6] and Zolotukhin, Nagaev, and Chebotarev [7] for the preceding bounds and historical context. The purpose of this paper is to lower the general upper bound from 0.4690 to 0.45.

The proof developed here has three layers. First, published estimates remove $n = 1$, large third moments, and $n \ge 8124$. Second, Prawitz's smoothing inequality is supplied with stable, distribution-free majorants. Third, the remaining compact parameter rectangle is covered by finitely many rational cells whose inequalities are recomputed with directed interval arithmetic. The logical interface between the last two layers is Proposition 6.3.

The analytic difficulty is concentrated in expressions that are harmless mathematically but ill-conditioned in floating-point arithmetic. Near $s = 0$, subtraction in $\cot(\pi s) - 1/(\pi s)$ is replaced by a positive series with an explicit tail. Near $u = 0$, subtraction between $\cos^n u$ and its Gaussian approximation is replaced by a product of nonnegative factors. Near $s = 1$, the Prawitz kernel is rewritten in the variable $h = 1-s$, exposing its linear zero and yielding a contribution of order H^2 on $[1 - H, 1]$.

The computer-assisted part is intentionally narrow. An A100 calculation supplies parameter advice, but no floating-point GPU value is trusted. The public checker parses exact rational data, reconstructs every partition, and recomputes every bound with outward rounding. It accepts only a complete, hash-bound bundle for all $n = 2, \ldots, 8123$. Thus search, parallel scheduling, and checkpoints affect running time, not the mathematical conclusion.

2 Reduction to a finite parameter rectangle

We use three established estimates. The sharp one-summand result of Bentkus and Kirša [1] gives

$$\Delta_1 \le 0.3704\,\rho < 0.45\,\rho. \tag{1}$$

A structural Berry–Esseen estimate due to Shevtsova [6] implies

$$\Delta_n \le \frac{0.3322(\rho + 0.429)}{\sqrt{n}}. \tag{2}$$

Finally, an asymptotically optimal estimate with an explicit quadratic remainder [5] gives

$$\Delta_n \le C_E \frac{\rho}{\sqrt{n}} + 3\left(\frac{\rho}{\sqrt{n}}\right)^2, \qquad C_E = \frac{\sqrt{10}+3}{6\sqrt{2\pi}}. \tag{3}$$

By Lyapunov's inequality, $\rho \ge 1$. Define

$$\rho_0 = \frac{0.3322 \cdot 0.429}{0.45 - 0.3322} = \frac{712569}{589000} = 1.2097945670\ldots. \tag{4}$$

Lemma 2.1 (Large third moments)**.** *If $\rho \ge \rho_0$, then*

$$\Delta_n \le 0.45\,\frac{\rho}{\sqrt{n}}$$

for every $n \ge 1$.

Proof. By (2), it is enough to check $0.3322(\rho + 0.429) \le 0.45\rho$, which is equivalent to $\rho \ge \rho_0$. □

Lemma 2.2 (Large sample sizes)**.** *If* $1 \le \rho \le \rho_0$ *and* $n \ge 8124$*, then*

$$\Delta_n \le 0.45\,\frac{\rho}{\sqrt{n}}.$$

Proof. Equation (3) gives

$$\frac{\sqrt{n}\,\Delta_n}{\rho} \le C_E + \frac{3\rho}{\sqrt{n}} \le C_E + \frac{3\rho_0}{\sqrt{n}}.$$

The right-hand side decreases with n. The strict inequality at $n = 8124$ follows from the exact rational calculation in Appendix A. □

Corollary 2.3 (Finite rectangle)**.** *To prove* $C^* \le 0.45$*, it is enough to treat*

$$2 \le n \le 8123, \qquad 1 \le \rho \le \rho_0. \tag{5}$$

Proof. Use (1) for $n = 1$, Lemma 2.1 for $\rho \ge \rho_0$, and Lemma 2.2 for $n \ge 8124$ in the remaining range. □

3 The Prawitz functional

Let

$$f_n(t) = \mathbb{E}e^{itS_n}, \qquad r_n(t) = |f_n(t) - e^{-t^2/2}|.$$

For $0 < s < 1$, define the Prawitz kernel

$$K(s) = \frac{1-s}{2} + \frac{\mathrm{i}}{2}\left((1-s)\cot(\pi s) + \frac{1}{\pi}\right), \tag{6}$$

and extend it continuously by $K(1) = 0$. Prawitz's smoothing inequality [4, 2] states that, for $n > 1$, $T > 0$, and $0 < t_0 \le 1$,

$$\begin{aligned}\Delta_n \le{}& 2\int_0^{t_0} |K(s)|\,r_n(Ts)\,\mathrm{d}s \\ &+ 2\int_{t_0}^{1} |K(s)|\,|f_n(Ts)|\,\mathrm{d}s \\ &+ 2\int_0^{t_0} \left|K(s) - \frac{\mathrm{i}}{2\pi s}\right| e^{-T^2s^2/2}\,\mathrm{d}s + \frac{1}{\pi}\int_{t_0}^{\infty} \frac{e^{-T^2s^2/2}}{s}\,\mathrm{d}s.\end{aligned} \tag{7}$$

The final term equals

$$\frac{1}{2\pi}E_1\left(\frac{T^2t_0^2}{2}\right), \qquad E_1(x) = \int_x^\infty \frac{e^{-y}}{y}\,\mathrm{d}y.$$

Set

$$B(\rho) = \sqrt{\frac{\rho^2}{2} + \frac{\rho\sqrt{\rho^2+8}}{2} - 2}. \tag{8}$$

Write $\varphi(u) = \mathbb{E}e^{iuY_1}$, so that $f_n(t) = \varphi(t/\sqrt{n})^n$. The comparison theorem of Mattner and Shevtsova [3] gives the one-summand estimate

$$|\varphi(u) - \cos u| \le \frac{B(\rho)|u|^3}{6}.$$

Put

$$u = \frac{t}{\sqrt{n}}, \qquad c = |\cos u|, \qquad \delta = \frac{B(\rho)|u|^3}{6}.$$

The power factorization, together with $|\varphi(u)| \le 1$ and $c \le 1$, first yields

$$\left| f_n(t) - \cos^n\left(\frac{t}{\sqrt{n}}\right) \right| \le n\delta = \frac{B(\rho)|t|^3}{6\sqrt{n}}. \tag{9}$$

Moreover, $|\varphi(u)| \le \min\{1, c+\delta\} =: a$. Define

$$D_{\mathrm{pow}}(n,\rho,t) = \delta \sum_{j=0}^{n-1} a^j c^{n-1-j}, \qquad D(n,\rho,t) = \min\{2, n\delta, D_{\mathrm{pow}}(n,\rho,t)\}.$$

The factorization of $z^n - w^n$, with $|\varphi(u)| \le a$, gives the bound D_{pow}. The triangle inequality gives the bound 2, while (9) gives $n\delta$. Taking the minimum gives

$$\left| f_n(t) - \cos^n\left(\frac{t}{\sqrt{n}}\right) \right| \le D(n,\rho,t). \tag{10}$$

Let θ_0 be the unique root in $(\pi, 2\pi)$ of

$$\theta^2 + 2\theta\sin\theta + 6(\cos\theta - 1) = 0,$$

and define

$$\kappa = \frac{\cos\theta_0 - 1 + \theta_0^2/2}{\theta_0^3}.$$

For $z \ge 0$, set

$$\psi(z) = \begin{cases} z^2/2 - \kappa z^3, & 0 \le z \le \theta_0, \\ 1 - \cos z, & \theta_0 < z \le 2\pi, \\ 0, & z > 2\pi. \end{cases} \tag{11}$$

Lemma 3.1 (Bounds for ψ)**.** *The function ψ in (11) is continuous on $[0,\infty)$ and satisfies*

$$0 \le \psi(z) \le 2 \qquad (z \ge 0).$$

Consequently, for $\rho \ge 1$, the base in (13) belongs to $[0,1]$.

Proof. Write

$$F(z) = z^2 + 2z\sin z + 6(\cos z - 1)$$

and, for $z > 0$,

$$r(z) = \frac{\cos z - 1 + z^2/2}{z^3}.$$

Direct differentiation gives

$$r'(z) = -\frac{F(z)}{2z^4}, \qquad F'(z) = 4\cos(z/2)\big(z\cos(z/2) - 2\sin(z/2)\big).$$

For $0 < z < \pi$, the inequality $\tan(z/2) > z/2$ implies $F'(z) < 0$, and hence $F(z) < 0$. For $\pi < z < 2\pi$, both factors in the displayed product for F' are negative, so $F'(z) > 0$. Moreover,

$$F(\pi) = \pi^2 - 12 < 0, \qquad F(2\pi) = 4\pi^2 > 0.$$

The intermediate value theorem therefore gives a zero in $(\pi, 2\pi)$, and strict monotonicity there makes it unique. It follows that $F(z) < 0$ for $0 < z < \theta_0$. Thus r is increasing there and

$$r(z) \le r(\theta_0) = \kappa \qquad (0 < z \le \theta_0).$$

Therefore

$$z^2/2 - \kappa z^3 \le 1 - \cos z \le 2.$$

Moreover, $1 - \cos\theta_0 < \theta_0^2/2$ gives $\kappa > 0$, and

$$\kappa\theta_0 = \frac{1}{2} - \frac{1 - \cos\theta_0}{\theta_0^2} \le \frac{1}{2}.$$

Hence

$$z^2/2 - \kappa z^3 = z^2(1/2 - \kappa z) \ge 0 \qquad (0 \le z \le \theta_0).$$

The bounds on the other branches are immediate. The defining identity for κ gives continuity at θ_0, and both adjacent branches vanish at 2π. Finally, $(\rho+1)^2 \ge 4$. □

For $\epsilon > 0$, write

$$\Psi(t, \epsilon) = \frac{\psi(\epsilon|t|)}{\epsilon^2}.$$

For $\delta = 1$, Theorem 2.2 of Shevtsova [5] states

$$|f_n(t)| \le \left(1 - \frac{2}{n}\Psi(t, \epsilon_n)\right)^{n/2}, \qquad \epsilon_n = \frac{\rho + \mathbb{E}|Y|}{\sqrt{n}}.$$

The cited result also proves that $\Psi(t, \epsilon)$ is nonincreasing in ϵ for fixed t. Since Cauchy–Schwarz gives $\mathbb{E}|Y| \le 1$, we may replace ϵ_n by the larger value $(\rho+1)/\sqrt{n}$. After the scaling

$$T = \frac{q\sqrt{n}}{\rho + 1}, \tag{12}$$

this gives

$$|f_n(Ts)| \le M_P(n, \rho, Ts) := \left(1 - \frac{2\psi(qs)}{(\rho+1)^2}\right)^{n/2}. \tag{13}$$

Indeed, $((\rho+1)/\sqrt{n})Ts = qs$, and hence $n^{-1}\Psi(Ts, (\rho+1)/\sqrt{n}) = \psi(qs)/(\rho+1)^2$, exactly the base in (13). Lemma 3.1 makes the real half-power in (13) well defined.

Define

$$R_{n,\rho}(t) = \min\left\{2, \left|\cos^n\left(\frac{t}{\sqrt{n}}\right) - \mathrm{e}^{-t^2/2}\right| + D(n, \rho, t)\right\}, \tag{14}$$

$$M_R(n, \rho, t) = \min\left\{1, \left|\cos\left(\frac{t}{\sqrt{n}}\right)\right|^n + D(n, \rho, t)\right\}, \tag{15}$$

and

$$M_{n,\rho}(t) = \min\{M_R(n, \rho, t), M_P(n, \rho, t)\}. \tag{16}$$

Then

$$r_n(t) \le R_{n,\rho}(t), \qquad |f_n(t)| \le M_{n,\rho}(t) \le 1. \tag{17}$$

Substitution into (7) gives

$$\begin{aligned} U_n(\rho; T, t_0) = {} & 2\int_0^{t_0} |K(s)| R_{n,\rho}(Ts)\,\mathrm{d}s \\ & + 2\int_{t_0}^{1} |K(s)| M_{n,\rho}(Ts)\,\mathrm{d}s \\ & + 2\int_0^{t_0} |L(s)| \mathrm{e}^{-T^2 s^2/2}\,\mathrm{d}s + \frac{1}{2\pi} E_1\left(\frac{T^2 t_0^2}{2}\right), \end{aligned} \tag{18}$$

where $L(s) = K(s) - \mathrm{i}/(2\pi s)$. For every admissible distribution,

$$\Delta_n \le U_n(\rho; T, t_0). \tag{19}$$

4 Stable analytic majorants

4.1 The coefficient $B(\rho)$

The direct formula (8) contains cancellation at $\rho = 1$. The following factorization removes it.

Lemma 4.1. *For every $\rho \ge 1$,*

$$2B(\rho)^2 = (\rho^2 - 1)\left(1 + \frac{\rho^2 + 9}{\rho\sqrt{\rho^2 + 8} + 3}\right). \tag{20}$$

Moreover, $B(\rho) \ge 0$ and B is increasing on $[1, \infty)$.

Proof. Rationalizing the radical in (8) gives (20). For $\rho > 1$,

$$\frac{\mathrm{d}}{\mathrm{d}\rho} B(\rho)^2 = \rho + \frac{\rho^2 + 4}{\sqrt{\rho^2 + 8}} > 0.$$

Since B is the nonnegative square root in (8) and $B(1) = 0$, monotonicity follows by continuity. □

4.2 A positive recursion for power differences

Lemma 4.2. *For $a, c \ge 0$ and every integer $m \ge 1$, let*

$$S_n(a, c) = \sum_{j=0}^{n-1} a^j c^{n-1-j}.$$

Then

$$S_{2m} = S_m(a, c)(a^m + c^m), \qquad S_{2m+1} = a^{2m} + cS_{2m}(a, c).$$

Proof. Split the first sum at m, and separate the term a^{2m} in the second. The recursion uses only addition and multiplication of nonnegative quantities and requires no division by $a - c$; it is therefore well suited to outward-rounded evaluation. □

4.3 The regularized cotangent

For $0 < s < 1$, define

$$d(s) = \frac{1}{\pi s} - \cot(\pi s). \tag{21}$$

Lemma 4.3 (Positive partial fractions)**.** *For $0 < x < \pi$,*

$$\frac{1}{x} - \cot x = 2x \sum_{k=1}^{\infty} \frac{1}{k^2\pi^2 - x^2}. \tag{22}$$

Every summand is positive and strictly increasing in x. For each integer $m \ge 1$,

$$\frac{1}{x} - \cot x \le 2x \sum_{k=1}^{m} \frac{1}{k^2\pi^2 - x^2} + \frac{2x}{m(\pi^2 - x^2/(m+1)^2)}. \tag{23}$$

Consequently, d is nonnegative and strictly increasing on $(0, 1)$.

Proof. The logarithmic derivative of Euler's product

$$\frac{\sin x}{x} = \prod_{k=1}^{\infty} \left(1 - \frac{x^2}{k^2\pi^2}\right),$$

with termwise differentiation justified uniformly on compact subsets of $(-\pi, \pi)$, gives (22). For fixed k,

$$\frac{\mathrm{d}}{\mathrm{d}x} \frac{x}{k^2\pi^2 - x^2} = \frac{k^2\pi^2 + x^2}{(k^2\pi^2 - x^2)^2} > 0.$$

For $k \geq m+1$,

$$\frac{1}{k^2\pi^2 - x^2} \leq \frac{1}{k^2(\pi^2 - x^2/(m+1)^2)}.$$

Using $\sum_{k=m+1}^{\infty} k^{-2} \leq \int_m^{\infty} t^{-2}\,\mathrm{d}t = 1/m$ proves (23). Substitution $x = \pi s$ proves the final assertion. □

The regularized kernel has the exact form

$$L(s) = \frac{1-s}{2}(1 - \mathrm{i}d(s)), \qquad 2|L(s)| = (1-s)\sqrt{1 + d(s)^2}. \tag{24}$$

Corollary 4.4 (Uniform L-bound)**.** *If $0 < s_- \leq s_+ < 1$, then*

$$\sup_{s \in [s_-, s_+]} 2|L(s)| \leq (1 - s_-)\sqrt{1 + d(s_+)^2}. \tag{25}$$

The value $d(s_+)$ may be replaced by the finite upper bound in (23).

Proof. Use (24), Lemma 4.3, and monotonicity of the square root. □

4.4 A stable Gaussian comparison

Directly subtracting $\cos^n u$ and $\mathrm{e}^{-nu^2/2}$ is poorly conditioned near zero. The next identity separates the small defect from a Gaussian factor.

Lemma 4.5 (Gaussian defect)**.** *Let $n \geq 1$ be an integer. For $0 \leq u < \pi/2$, define*

$$h(u) = -\log(\cos u) - \frac{u^2}{2}. \tag{26}$$

Then $h(0) = 0$, $h(u) \geq 0$, and h is increasing. Moreover,

$$\left|\cos^n u - \mathrm{e}^{-nu^2/2}\right| = \mathrm{e}^{-nu^2/2}\left(1 - \mathrm{e}^{-nh(u)}\right). \tag{27}$$

Proof. Since $h'(u) = \tan u - u$ and $(\tan u - u)' = \tan^2 u \geq 0$, one has $h' \geq 0$ and $h \geq 0$. Positivity of $\cos u$ permits taking the logarithm and gives

$$\cos^n u = \mathrm{e}^{-nu^2/2}\mathrm{e}^{-nh(u)}.$$

Equation (27) follows. □

Corollary 4.6 (Rectangular-cell bound)**.** *Let $n \geq 1$ be an integer, $\rho \in [\rho_-, \rho_+]$, $s \in [s_-, s_+]$, and $q > 0$. Put*

$$u_- = \frac{qs_-}{\rho_+ + 1}, \qquad u_+ = \frac{qs_+}{\rho_- + 1}.$$

If $0 \leq u_- \leq u_+ < \pi/2$, then throughout the cell

$$\begin{aligned} \left|\cos^n\left(\frac{qs}{\rho+1}\right) - \exp\left(-\frac{nq^2s^2}{2(\rho+1)^2}\right)\right| &\leq \mathrm{e}^{-nu_-^2/2}\left(1 - \mathrm{e}^{-nh(u_+)}\right) \\ &\leq \mathrm{e}^{-nu_-^2/2}\min\{1, nh(u_+)\}. \end{aligned} \tag{28}$$

Proof. Throughout the rectangle,

$$u_- \le \frac{qs}{\rho+1} \le u_+.$$

In (27), the Gaussian factor decreases with u, while $1-\mathrm{e}^{-nh(u)}$ increases. Bound the two factors at opposite endpoints. The final inequality uses $1-\mathrm{e}^{-x} \le \min\{1,x\}$ for $x \ge 0$. □

5 Analytic control of the endpoint $s = 1$

The second integral in (18) reaches $s = 1$. A rectangular enclosure that forgets $K(1) = 0$ introduces a first-order artifact. The endpoint can instead be handled exactly.

Lemma 5.1 (Endpoint identity)**.** *For $0 < h < 1$,*

$$K(1-h) = \frac{h}{2} + \frac{\mathrm{i}h}{2} d(h), \qquad 2|K(1-h)| = h\sqrt{1+d(h)^2}. \tag{29}$$

In particular, $K(1) = 0$ by continuity.

Proof. Using $\cot(\pi - \pi h) = -\cot(\pi h)$ in (6) gives

$$K(1-h) = \frac{h}{2} + \frac{\mathrm{i}}{2}\left(-h\cot(\pi h) + \frac{1}{\pi}\right) = \frac{h}{2} + \frac{\mathrm{i}h}{2} d(h).$$

Taking the modulus proves (29). Equation (22) gives $d(h) = O(h)$ as $h \downarrow 0$, and hence $K(1-h) \to 0$. □

Lemma 5.2 (Uniform endpoint shape)**.** *If $0 \le h \le H \le 1/2$, then*

$$2|K(1-h)| \le h\sqrt{1+d(H)^2}. \tag{30}$$

Proof. For $h = 0$, the assertion follows from $K(1) = 0$. If $0 < h \le H$, Lemma 4.3 gives $0 \le d(h) \le d(H)$; now apply (29). □

Theorem 5.3 (Quadratic endpoint tail)**.** *Let $n \ge 1$ be an integer, let $\rho \ge 1$, $T > 0$, and let $0 < H \le 1/2$, $s_* = 1 - H$. Then*

$$\int_{s_*}^{1} 2|K(s)| M_{n,\rho}(Ts)\,\mathrm{d}s \le \frac{H^2}{2}\sqrt{1+d(H)^2}. \tag{31}$$

Proof. By (17), $0 \le M_{n,\rho} \le 1$. Substitute $s = 1 - h$, apply Lemma 5.2, and integrate:

$$\int_{s_*}^{1} 2|K(s)| M_{n,\rho}(Ts)\,\mathrm{d}s \le \int_0^H h\sqrt{1+d(H)^2}\,\mathrm{d}h.$$

□

Remark 5.4 (Sharp order). Equation (22) gives

$$d(H) = \frac{\pi H}{3} + O(H^3), \qquad \frac{H^2}{2}\sqrt{1+d(H)^2} = \frac{H^2}{2} + O(H^4).$$

A bound that treats the endpoint imaginary part as a nonzero constant would produce an artificial $O(H)$ contribution. The quadratic estimate restores the correct local order of the kernel.

Lemma 5.5 (Choice of cutoff)**.** *Let $r \geq 1$ be an integer and let*

$$\max\{t_0, 1/2\} \leq s_1, \ldots, s_r < 1.$$

For each j, suppose that P_j is a rigorous upper bound for

$$\int_{t_0}^{s_j} 2|K(s)|M_{n,\rho}(Ts)\,\mathrm{d}s.$$

Then

$$\int_{t_0}^{1} 2|K(s)|M_{n,\rho}(Ts)\,\mathrm{d}s \leq \min_{1\leq j\leq r}\left\{P_j + \frac{(1-s_j)^2}{2}\sqrt{1+d(1-s_j)^2}\right\}. \tag{32}$$

Proof. For each j, split the integral at s_j. Since $H_j = 1 - s_j \in (0, 1/2]$, Theorem 5.3 applies to the second part. Every expression inside the minimum is therefore an upper bound for the complete integral, and the finite minimum is valid. □

Remark 5.6 (Verifier cutoffs). The verifier uses the exact dyadic cutoffs $s_j = 1 - 2^{-j}$ for $j = 1, \ldots, 14$, retaining only eligible cutoffs. It also tests the left endpoint of the final certificate subinterval when that endpoint is at least $1/2$. The value of d is bounded by (23) with $m = 100$.

6 A finite verification criterion

Fix $n \in \{2, \ldots, 8123\}$ and $I = [\rho_-, \rho_+] \subset [1, \rho_0]$. Choose positive rational q, t_0, η, λ such that

$$0 < \lambda \leq \min\{t_0, 1/2\}, \qquad t_0 < 1, \qquad 0 < \eta \leq \min\{t_0, 1/2\}, \qquad \frac{q\eta}{\rho_- + 1} \leq 1,$$

and use $T(\rho) = q\sqrt{n}/(\rho+1)$ as in (12). For every $\rho \in I$,

$$\frac{T(\rho)\eta}{\sqrt{n}} = \frac{q\eta}{\rho+1} \leq \frac{q\eta}{\rho_-+1} \leq 1.$$

Lemma 6.1 (First integral near zero)**.** *Let $n \geq 1$ be an integer, $\rho \geq 1$, $T > 0$, and $0 < \eta \leq 1/2$. If $T\eta/\sqrt{n} \leq 1$, then*

$$2\int_0^{\eta} |K(s)|R_{n,\rho}(Ts)\,\mathrm{d}s \leq \frac{B(\rho)T^3\eta^3}{9\sqrt{n}} + \frac{T^4\eta^4}{12n}. \tag{33}$$

Proof. First justify the kernel estimate. If $0 < s < 1/2$, then $\tan(\pi s) \geq \pi s$, so $0 \leq \cot(\pi s) \leq 1/(\pi s)$. Equation (6) gives

$$|\Re K(s)| \leq \frac{1}{2}, \qquad |\Im K(s)| \leq \frac{1}{2\pi s}.$$

Thus

$$|K(s)| \leq |\Re K(s)| + |\Im K(s)| \leq \frac{1}{2} + \frac{1}{2\pi s} \leq \frac{1}{s}.$$

At $s = 1/2$, the same estimate follows directly from $\cot(\pi/2) = 0$; the endpoint $s = 0$ is understood by the displayed majorant.

Put $u = Ts/\sqrt{n}$. On $0 \leq s \leq \eta$, one has $0 \leq u \leq 1$. Lemma 4.5 gives $\cos u \leq \mathrm{e}^{-u^2/2}$. Moreover,

$$\cos u \geq 1 - \frac{u^2}{2}, \qquad \mathrm{e}^{-u^2/2} \leq 1 - \frac{u^2}{2} + \frac{u^4}{8},$$

and hence

$$0 \le \mathrm{e}^{-u^2/2} - \cos u \le \frac{u^4}{8}.$$

Both factors lie in $[0,1]$. The power factorization therefore gives

$$\left|\mathrm{e}^{-nu^2/2} - \cos^n u\right| \le n\left(\mathrm{e}^{-u^2/2} - \cos u\right) \le \frac{nu^4}{8} \le \frac{nu^4}{6}.$$

From the definition preceding (10),

$$D(n,\rho,Ts) \le n\delta = \frac{B(\rho)T^3s^3}{6\sqrt{n}}.$$

Equation (14) now gives

$$R_{n,\rho}(Ts) \le \frac{B(\rho)T^3s^3}{6\sqrt{n}} + \frac{T^4s^4}{6n}.$$

Multiply by $2|K(s)| \le 2/s$ and integrate. □

Lemma 6.2 (Third integral near zero)**.** *For* $0 < \lambda \le \min\{t_0, 1/2\}$,

$$2\int_0^\lambda |L(s)|\mathrm{e}^{-T^2s^2/2}\,\mathrm{d}s \le \lambda\sqrt{1+\frac{4}{\pi^2}}. \tag{34}$$

Proof. For $0 < s \le \lambda \le 1/2$, Lemma 4.3 gives $d(s) \le d(1/2) = 2/\pi$. Equation (24) implies

$$2|L(s)| = (1-s)\sqrt{1+d(s)^2} \le \sqrt{1+4/\pi^2}.$$

Since $\mathrm{e}^{-T^2s^2/2} \le 1$, integration proves (34); the endpoint $s = 0$ follows by continuity. □

6.1 Uniform evaluation on a cell

Partition $[\eta, t_0]$, $[t_0, 1]$, and $[\lambda, t_0]$ into finitely many rational subintervals. On every proper subrectangle $I \times [s_-, s_+]$, all quantities are enclosed by outward-rounded interval arithmetic.

Lemma 4.1 bounds $B(\rho)$ by $B(\rho_+)$. The scaled variable satisfies

$$\frac{qs_-}{\rho_+ + 1} \le \frac{qs}{\rho + 1} \le \frac{qs_+}{\rho_- + 1}.$$

Equations (8)–(17), Lemma 4.2, and Corollary 4.6 give uniform upper bounds for $R_{n,\rho}$ and $M_{n,\rho}$. If the upper endpoint of the scaled interval is not below $\pi/2$, the difference in (14) is evaluated directly by interval arithmetic. Definition (11) is evaluated on cells contained in one branch; a subinterval crossing θ_0 or 2π uses the global lower bound $\psi \ge 0$ from Lemma 3.1. Continuity of ψ at both branch points and continuity of the other elementary expressions show that the proper integrands are continuous.

Equation (6) bounds the kernel in the first two proper integrals. Equations (23)–(25) bound the regularized kernel in the third. Each proper integral is enclosed by a finite upper sum. For the second integral, every prefix sum is combined with Theorem 5.3 and Lemma 5.5.

The near-zero part of the first integral is bounded uniformly by (33), using $B(\rho_+)$ and

$$T_{\max} = \frac{q\sqrt{n}}{\rho_- + 1}.$$

The third near-zero part is bounded by (34). Finally,

$$E_1'(x) = -\frac{\mathrm{e}^{-x}}{x} < 0.$$

Since $T(\rho)$ is smallest at $\rho = \rho_+$, the last term in (18) is bounded uniformly by

$$\frac{1}{2\pi} E_1\left(\frac{nq^2t_0^2}{2(\rho_+ + 1)^2}\right).$$

Let $\widehat{U}_{n,I}$ denote the sum of these six certified contributions. Then

$$U_n(\rho; T(\rho), t_0) \le \widehat{U}_{n,I} \qquad (\rho \in I).$$

The implementation isolates the two branch points with the rational guards

$$\frac{3995895679}{10^9} < \theta_0 < \frac{3995895680}{10^9}, \qquad \frac{6283185307}{10^9} < 2\pi < \frac{6283185308}{10^9}.$$

These inequalities are themselves certified at initialization.

Proposition 6.3 (Cell criterion)**.** *If*

$$\frac{\sqrt{n}}{\rho_-}\,\widehat{U}_{n,I} \le \frac{9}{20}, \tag{35}$$

then

$$\frac{\sqrt{n}\,\Delta_n}{\rho} \le 0.45$$

for every $\rho \in I$ and every admissible distribution with third absolute moment ρ.

Proof. By (19), $\Delta_n \le \widehat{U}_{n,I}$ throughout the cell. Since $\rho \ge \rho_-$,

$$\frac{\sqrt{n}\,\Delta_n}{\rho} \le \frac{\sqrt{n}\,\widehat{U}_{n,I}}{\rho_-} \le \frac{9}{20} = 0.45.$$

□

Theorem 6.4 (Finite completion criterion)**.** *Suppose that, for every $n = 2, \dots, 8123$, the interval $[1, \rho_0]$ is covered by finitely many cells satisfying (35). Then $C^* \le 0.45$.*

Proof. Apply Proposition 6.3 on the finite rectangle (5), and use Corollary 2.3 outside it. □

7 Computer-assisted verification

The finite covering in Theorem 6.4 is implemented as a certificate-checking computation. The proof object is separated from the procedure used to discover it: GPU searches and heuristic parameter selection may propose a certificate, but acceptance depends on a compact checker that reconstructs every partition and recomputes every bound with outward-rounded interval arithmetic.

7.1 Scope and proof objects

For each integer $n = 2, \dots, 8123$, a certificate shard contains an ordered partition

$$1 = \rho_{n,0} < \rho_{n,1} < \cdots < \rho_{n,m_n} = \rho_0.$$

Each cell $I_{n,j} = [\rho_{n,j-1}, \rho_{n,j}]$ carries exact rational parameters q, t_0, η, λ and a compact grid recipe with two integers, `pieces` and `power`. The checker requires literal equality of consecutive rational endpoints. Gaps, overlaps, reversed intervals, and degenerate cells are rejected.

For each cell the checker reconstructs all six contributions to $\widehat{U}_{n,I}$: the two near-zero estimates (33) and (34), three proper upper sums, and the E_1 tail. It accepts the cell only if the outward-rounded upper endpoint of the left-hand side of (35) does not exceed the lower endpoint of the exact rational 9/20.

7.2 Exact certificate language

A shard has format `berry-esseen-045-cert-v2`. All real-valued certificate data are canonical rational strings: an integer or a reduced fraction a/b with $b > 0$. JSON floating-point numbers, duplicate keys, unknown fields, noncanonical fractions, and nonstandard constants such as `NaN` or `Infinity` are rejected. A cell contains exactly

$$\texttt{rho},\ \texttt{q},\ \texttt{t0},\ \texttt{eta},\ \texttt{lambda},\ \texttt{grid},$$

where `rho` contains two endpoints and $\texttt{grid} = \{\texttt{pieces}, \texttt{power}\}$, with

$$1 \le \texttt{pieces} \le 4096, \qquad 1 \le \texttt{power} \le 8.$$

The analytic guards

$$0 < \lambda \le \min\{t_0, 1/2\}, \qquad t_0 < 1, \qquad 0 < \eta \le \min\{t_0, 1/2\}, \qquad q\eta \le \rho_- + 1$$

are checked again from the parsed rationals.

7.3 Deterministic reconstruction of the grids

No floating-point grid nodes occur in the proof object. Let $[a, b]$ be one of the proper integration intervals, $p = \texttt{pieces}$, and $d = \texttt{power}$. A concentrated grid is the sorted union of the following exact rational points:

(i) the polynomial mesh

$$a + (b - a)\frac{i^d}{M^d}, \qquad M = \max\{8, \lfloor p/2 \rfloor\}, \qquad 0 \le i \le M;$$

(ii) the uniform safety mesh with $\max\{4, \lfloor p/8 \rfloor\}$ subintervals; and

(iii) for $n \ge 16$, a mesh in the scaled variable Ts. Writing

$$\underline{\sqrt{n}} = \frac{\lfloor 10^6 \sqrt{n} \rfloor}{10^6}, \qquad D = \max\{8, \lfloor p/4 \rfloor\},$$

where the floor is computed by integer square root, the inserted points are

$$\frac{\rho_{\mathrm{mid}} + 1}{q\,\underline{\sqrt{n}}}\frac{k}{D}, \qquad 0 \le k \le 10D,$$

whenever they lie strictly inside (a, b).

For $n < 16$ and $d = 1$, the concentrated grid reduces to the exact uniform p-partition. The first and third proper integrals use concentrated grids; the second uses the exact uniform p-partition. Every grid is augmented with the four rational guards for θ_0/q and $2\pi/q$ displayed in Section 6. Thus branch crossings of $\psi(qs)$ are isolated deterministically. The second grid is also augmented by the dyadic endpoint cutoffs described in Remark 5.6.

7.4 Outward-rounded evaluation

The trusted numerical core uses Arb through `python-flint==0.9.0`. Exact rational points are passed through FLINT rationals; an interval $[a, b]$ is represented by the ball with exact midpoint $(a + b)/2$ and radius $(b - a)/2$. The publication configuration uses 192 bits of precision. All elementary and special-function evaluations are outward rounded.

The decision path is fail-closed. Every computed enclosure must be finite. Denominators and square-root domains must be certified positive, the exponential-integral argument must be certified positive, and branch guards must be decisive. An ambiguous comparison is a rejection. The code uses the stable factorization (20), the positive recursion of Lemma 4.2, the cotangent upper bound (23) with 100 terms, and Corollary 4.6; direct interval subtraction is only the guarded fallback outside the positive-cosine region.

The endpoint directions in Section 6 are enforced explicitly: B is evaluated at ρ_+, the largest T at ρ_-, and the E_1 term from the smallest T, which occurs at ρ_+. The second integral tests several complete upper bounds from (32); taking their finite minimum is sound because no candidate is merely a bound for a partial tail.

Each worker initializes the constants independently. Initialization certifies the isolating bracket for θ_0, the rational bracket for 2π, positivity of the required constants, and the global reduction

$$C_E + \frac{3\rho_0}{\sqrt{8124}} < \frac{9}{20}.$$

7.5 Bundle completeness and integrity

A complete proof object has format `berry-esseen-045-certificate-bundle-v2`. It contains exactly 8122 ordered shards with the canonical names

$$\texttt{n/0002.json}, \ldots, \texttt{n/8123.json}.$$

The checker rejects an incorrect scope, missing or additional files, directories among the shards, renamed files, path traversal, or an incorrect order.

The manifest records the byte length and SHA–256 digest of each shard. A shard is read once; both the parsed object and its digest are derived from the same byte string. The checker also recomputes the ordered certificate-set digest obtained by hashing the ASCII records

$$\texttt{2:hash_2\textbackslash n}, \ldots, \texttt{8123:hash_8123\textbackslash n}.$$

The manifest records and the checker recomputes SHA–256 digests for the numerical core, the public checker, the constructor, the parameter-bank loader, the one-shot launcher, and both relevant package initializers. The parameter-bank manifest digest is retained as construction provenance. The constructor and bank remain advice-producing components rather than premises for a cell inequality: every proposed cell is recomputed by the checker. These content hashes detect inconsistency within a bundle; they do not authenticate the repository revision or replace review of the trusted source.

The complete checker performs the following ordered obligations.

1. Require the specified `python-flint` and underlying FLINT versions, parse the manifest strictly, and require the exact public scope.
2. Validate all 8122 entries, source hashes, file names, byte lengths, shard hashes, the certificate-set digest, and the absence of extra shards.
3. Initialize interval constants and verify the analytic large-n reduction.
4. For each n, require an exact gap-free cover of $[1, \rho_0]$, reconstruct every grid, and recompute every cell.
5. Only after all shards have returned in the canonical order, atomically write the bundle-local audit record and print `ACCEPTED_FULL`.

Before work begins, the one-shot launcher invalidates any previous accepted audit. The checker permits an audit only at the fixed bundle-local path, rejects symbolic or multiply linked targets, and installs the new record by atomic replacement. Every validation or arithmetic failure handled by the checker prints `REJECTED`, returns a nonzero exit status, and leaves no active accepted global result. Command-line parsing errors and an external process interruption also return nonzero, but are reported by the Python runtime rather than by that validation handler. A diagnostic single-shard acceptance is not a certificate of the theorem.

7.6 Trust boundary

Table 1: Logical role of the software and data.

Component	Role	Trusted
Analytic reductions and lemmas in this paper	Mathematical derivation of the finite criterion	yes
`proof/berry_core.py`	Outward-rounded realization of equations (20)–(35)	yes
`proof/check_certificate.py`	Strict parsing, coverage, integrity, and complete recomputation	yes
Package initializers and `construction/verify.py`	Faithful one-shot launch, manifest assembly, and result reporting	execution
CPython, `python-flint`, FLINT, and Arb	Runtime and interval-arithmetic substrate	yes
Linux kernel, Docker engine, CPU, and hardware	Host execution substrate outside the container image	yes
Certificate constructor and parameter-bank loader	Candidate selection, refinement, and checkpointing	no
A100 parameter bank and GPU notebooks	Floating-point advice used to find efficient rational recipes	no
Printed decimals, elapsed time, and worst-cell summaries	Human-readable audit information	no

The trusted numerical core and checker form the mathematical project-specific surface. The launcher and package initialization path must also report faithfully when the one-shot command is used; their exact source bytes are bound into the manifest. Runtime libraries are execution assumptions and are pinned by version and binary hashes in the supplied container. The host kernel, container engine, processor, and hardware remain ordinary external execution assumptions: a container fixes userspace, not those components. The

final checker is launched in a fresh process after construction. This separates acceptance from constructor state, but it is not an implementation-diverse second realization of the analytic formulas.

7.7 Untrusted construction and reproducibility

The parameter bank contains 886,998 dyadic ρ-cells and six candidate recipes per cell. The exact geometry is recovered from integer depth and index data:

$$\left[1 + \frac{j}{2^d}(\rho_0 - 1),\ 1 + \frac{j+1}{2^d}(\rho_0 - 1)\right].$$

The bank coordinates for candidate q, t_0, η, λ begin on the dyadic lattice with denominator 2^{20}. The constructor converts them to exact rationals and may raise η to the exact admissibility boundary $(\rho_- + 1)/q$; consequently, a parameter stored in the final certificate need not retain denominator 2^{20}. The constructor tries the six candidates in a fixed order and, if necessary, doubles the proposed number of grid pieces up to 4096. Every emitted candidate is already evaluated with Arb, but that constructor-side test is an optimization only; the public checker recomputes the cell.

Construction is deterministic for fixed sources, configuration, bank, and runtime. Each n is handled in a separate process with no cross-n warm start. Parallelism changes scheduling only. A completion marker binds n, the mathematical configuration digest, the imported construction-source digest, shard digest, and byte length. Legacy markers from the preceding format are accepted only while the previously omitted imported sources still equal their frozen release hashes. Shards and markers are installed by atomic replacement, so interruption cannot promote a partial file. A resumed run reuses only markers whose content and hashes still match. The launcher holds an exclusive operating-system lock on the artifact tree for the entire construction and checking sequence, preventing simultaneous runs from racing on shards, markers, logs, or the final audit.

The complete Windows execution is:

```
verify.cmd
```

It builds the pinned Linux container, constructs or resumes all shards, and launches a fresh complete check. The default is six worker processes; output reports every n through 10 and then every multiple of ten. The build targets a pinned `linux/amd64` userspace and fixes Python 3.12.4, the Docker frontend and base-image digests, and the hashes of all installed wheels. The bank is mounted read-only, project artifacts are the only writable output, network access is disabled, Linux capabilities are dropped, and the verifier runs as a non-root user.

Regeneration of the advice bank is a separate reproducibility experiment. The output-free notebook `notebooks/build_parameter_bank_colab.ipynb` checks out an immutable 40-hex-character Git commit and requires CPython 3.12 on Linux x86-64 and an NVIDIA A100 of compute capability 8.0. The complete Python wheel dependency closure is version- and SHA–256-locked and forcibly reinstalled; the Colab-provided CUDA runtime is required to be 12.x, while the exact runtime and driver versions are recorded rather than fixed by that wheel lock. Checkpoints bind the build profile, exact input geometry, parent payload, and saved payload. A completed bank is copied to a content-addressed Drive directory and exposed through an atomically installed pointer only after the copied tree is authenticated. The builder publishes a bank only after array shapes, dtypes, byte counts, SHA–256 digests, exact dyadic coverage, and semantic equality with the committed reference manifest; it then installs the exact reference manifest bytes. An A100 is therefore needed to reproduce the search output, but not to verify a completed certificate.

Proposition 7.1 (Checker implication)**.** *Assume the analytic statements used by the trusted numerical core and faithful execution of the stated trusted stack. If the public checker terminates with* `ACCEPTED_FULL`*, then the hypothesis of Theorem 6.4 has been verified for every* $n = 2, \dots, 8123$ *and every* $\rho \in [1, \rho_0]$*. Consequently,*

$$C^* \le \frac{9}{20} = 0.45.$$

Proof. Acceptance implies that every required n occurs exactly once and that its cells form an exact cover of $[1, \rho_0]$. On each cell, outward-rounded evaluation proves (35). Their union gives the hypothesis of Theorem 6.4 throughout the finite rectangle. Initialization checks the threshold $n = 8124$, and Section 2 covers all complementary ranges. □

8 Exploratory construction of the parameter bank

The parameter advice was obtained by a float64 search on an NVIDIA A100-SXM4 80 GB GPU. It determines where fine ρ-subdivision and dense integration meshes are useful. Its sampled values are not consumed by the public checker.

The final bank contains 886,998 leaves. Its exact geometry is summarized in Table 2; the geometry digest is

`bcbe1ee1f583595a929a55043b10e1637a43cc1c76931eddad621c5ef8a3f849`.

Table 2: Final dyadic geometry of the publication parameter bank.

Depth	Final leaves	Percentage
6	430,226	48.5036%
8	259,884	29.2993%
9	196,888	22.1971%
Total	886,998	100%

In the exploratory scoring used to design the bank, the distribution of the largest sampled score within a cell was as follows.

Table 3: Exploratory float64 scores. These values are search diagnostics, not interval bounds.

Sampled normalized score	Cells	Percentage
$B_{\mathrm{num}} < 0.4400$	886,076	99.8961%
$0.4400 \le B_{\mathrm{num}} < 0.4450$	361	0.0407%
$0.4450 \le B_{\mathrm{num}} < 0.4480$	467	0.0526%
$0.4480 \le B_{\mathrm{num}} < 0.4495$	94	0.0106%
$B_{\mathrm{num}} \ge 0.4495$	0	0%

The largest sampled score was

$$B_{\mathrm{num}}^{\max} = 0.44820233914501284, \tag{36}$$

with the difficult region concentrated near small n, in particular near $n = 3$. The gap to 0.45 guided the grid design but has no logical role in acceptance.

Checkpoint restoration was used during the bank search. Consequently, the recorded 1914.5-second active segment in the exploratory notebook is not an end-to-end reproduction time and is not reported as one. A fresh reproduction should report wall-clock time together with which levels were restored; the complete certificate checker separately records its own elapsed time in the final audit.

9 Discussion

The endpoint estimate in Theorem 5.3 changes the geometry of the verification problem. A rectangular bound that forgets $K(1) = 0$ forces the final mesh width to compensate for an $O(H)$ artifact. The regularized identity gives the correct $O(H^2)$ contribution, allowing much of the second Prawitz integral to be certified without an excessively fine endpoint grid.

The partial-fraction expansion and Gaussian-defect identity serve the same purpose at the other delicate points: they replace subtraction of nearly equal quantities by positive expressions. Positivity makes monotonicity transparent and prevents interval widths from growing merely because two large overlapping enclosures are subtracted.

The method also illustrates the useful separation between discovery and verification. A large GPU search can optimize parameters aggressively without entering the trusted base. Once a rational candidate is serialized, the public checker discards every exploratory score and reconstructs the mathematical inequality from first principles. A defective parameter bank can slow construction or prevent it from finding a certificate; it cannot cause an invalid cell to pass.

Data and code availability

The complete checker, trusted numerical core, certificate schemas, container specification, parameter-bank provenance, and one-command reproduction instructions are supplied in the accompanying public repository. The certificate shards contain only exact rational data. The accepted audit record is written to `artifacts/certificate/verification.json`; the executable checker, not the human-readable decimal summaries in that record, is authoritative.

A Exact verification of the threshold $n = 8124$

Set

$$a = \frac{3162277661}{10^9}, \qquad p = \frac{3141592653}{10^9}, \qquad b = \frac{2506628274}{10^9}, \qquad c = \frac{90133234}{10^6}.$$

For completeness, define

$$A_m(x) = \sum_{k=0}^{m} \frac{(-1)^k x^{2k+1}}{2k+1}.$$

The alternating-series remainder gives $A_{2r+1}(x) < \arctan x < A_{2r}(x)$ for $0 < x \le 1$. Machin's identity $\pi = 16\arctan(1/5) - 4\arctan(1/239)$ therefore yields

$$\pi > L := 16A_7(1/5) - 4A_2(1/239) = \frac{3367720447004809249690544948}{1071978712185331878662109375}.$$

The entirely rational comparison

$$L - p = \frac{323056151203315029001}{54885310063888992187500000000} > 0$$

proves $p < \pi$. Direct integer arithmetic gives

$$a^2 - 10 = \frac{5259630921}{10^{18}} > 0,$$

$$2p - b^2 = \frac{495945231}{250000000000000000} > 0,$$

$$8124 - c^2 = \frac{32175311}{250000000000} > 0,$$

and

$$\frac{409732184}{10^9} - \frac{a+3}{6b} = \frac{65406953}{469992801375000000} > 0.$$

Consequently,

$$\sqrt{10} < a, \qquad \sqrt{2\pi} > b, \qquad \sqrt{8124} > c, \qquad C_E < \frac{409732184}{10^9}.$$

Using $\rho_0 = 712569/589000$, one more exact subtraction yields

$$\frac{9}{20} - \left(\frac{409732184}{10^9} + \frac{3\rho_0}{90133234/10^6}\right) = \frac{3121000875001}{3318029676625000000} > 0.$$

This proves $C_E + 3\rho_0/\sqrt{8124} < 9/20$ without floating-point evaluation.

B Correspondence between the manuscript and the trusted core

Table 4 identifies the principal executable realizations of the analytic statements. It is a review map, not an additional assumption.

Table 4: Analytic-to-executable correspondence.

Trusted function	Mathematical result
`stable_B_point`	Lemma 4.1 and equation (20)
`geom_sum_upper`	Lemma 4.2
`regularized_cot_d_upper_point`	Lemma 4.3 and equations (21)–(23)
`kernel_L_abs_upper`	Corollary 4.4 and equation (25)
`rad_gauss_diff_upper`	Lemma 4.5, Corollary 4.6, and equations (26)–(28)
`kernel_K_abs_upper`	Lemmas 5.1–5.2 and equations (29)–(30)
`second_endpoint_integral_upper`	Theorem 5.3 and equation (31)
`second_integral_upper`	Lemma 5.5 and equation (32), with $s_* \geq 1/2$
`evaluate_cell`	Lemmas 6.1–6.2, Proposition 6.3, and equations (33)–(35)
`self_test_global_reductions`	Lemma 2.2 and Appendix A

Certified conclusion

Reproducibility record. **Archival verification.**

The public checker returned `ACCEPTED_FULL` for bundle format `berry-esseen-045-certificate-bundle-v2`, target $9/20$, and exact scope $n = 2, \ldots, 8123$. It

completed on 2026-08-06T20:09:07Z using
`Linux-6.6.87.2-microsoft-standard-WSL2-x86_64-with-glibc2.36`, `python-flint` 0.9.0, and FLINT 3.6.0; the complete check took 38558.206 seconds and accepted 886998 cells. The largest reported directed-decimal cell upper bound was 0.44999851443545250668104334952 2 at $n = 13$, cell 21, on $\rho \in [304162949/301568000, 152143259/150784000]$.
The SHA–256 digest of the accepted manifest is

`0fb539f0341492181a2eb23d0b2a512cab50a819a2582c955f24cb5dd22a24f5`

It binds the complete certificate-set digest and every packaged source digest; the audit record lists the bound digests separately.

Theorem. For every $n \geq 1$ and every independent identically distributed sequence with mean zero, variance $\sigma^2 > 0$, and finite third absolute moment,

$$\sup_{x\in\mathbb{R}} \left| \mathbb{P}\left\{ \frac{X_1 + \cdots + X_n}{\sigma\sqrt{n}} \leq x \right\} - \Phi(x) \right| \leq \frac{9}{20} \frac{\mathbb{E}|X_1|^3}{\sigma^3\sqrt{n}}.$$

In particular, the universal Berry–Esseen constant satisfies

$$\boxed{C^* \leq \frac{9}{20} = 0.45}.$$

Proof. The archival bundle-v2 audit records `ACCEPTED_FULL` for target 9/20 and the exact scope $n = 2, \ldots, 8123$, after checking the manifest, certificate-set, and packaged-source digests. Proposition 7.1 proves the finite rectangle, and Corollary 2.3 proves every complementary case. □

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